PROJECT REPORT ON

MultiModal Visual Medical Diagnosis Assistant Using Deep Learning and Retrieval-Augmented Clinical Knowledge

(LAB X)

Submitted in partial fulfilment
for the award of the Course of

Machine Learning to Generative AI and LLMs

Centre for Development of Advanced Computing (C-DAC)
Mohali, India

Guided By:

Evaluation Board
C-DAC Mohali

Submitted By:

Raghav Sarwan (250811050041)
Anika (250811050042)

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to all those who have supported and guided me throughout the course of my major project titled **'MultiModal Visual Medical Diagnosis Assistant Using Deep Learning and Retrieval-Augmented Clinical Knowledge' (LAB X)**. Without their contributions, academic insights, and constant encouragement, the successful completion of this project would not have been possible.

Firstly, I would like to extend my heartfelt thanks to our supervisors at **C-DAC Mohali** for their invaluable guidance, continuous support, and technical insights. Their expertise in machine learning and data engineering was instrumental in helping me overcome the architectural challenges of integrating deep learning computer vision with semantic retrieval databases.

I am also deeply grateful to **C-DAC Mohali** for providing the necessary academic resources and a highly conducive environment for research. Lastly, I would like to thank my project partner **Anika** for her incredible dedication to coding the RAG databases and Streamlit dashboard integrations, which made this system a success.

Raghav Sarwan
Anika
Machine Learning to Generative AI and LLMs

ABSTRACT

With the increasing volume of medical imaging scans (MRIs, CTs, X-Rays) in modern healthcare, rapid and accurate radiological interpretation has become a critical bottleneck. Standard computer vision models operate as 'black boxes,' offering diagnostic classifications without explaining the spatial reasons behind their decisions. Furthermore, clinicians often need to cross-reference diagnoses with scattered clinical guidelines and drug protocols, which is time-consuming.

This project presents **LAB X**, an explainable Multimodal Visual Medical Diagnosis Assistant. The system utilizes a pre-trained **EfficientNet-B4** deep learning backbone in PyTorch to classify pathologies across four modalities: **Chest X-Rays, Brain MRIs, Abdominal CTs, and Knee MRIs**. To solve the black-box challenge, we implement a localized **Grad-CAM (Gradient-weighted Class Activation Mapping)** engine that overlays 2D Gaussian attention heatmaps at specific anatomical coordinates on the scans.

Simultaneously, we construct a secure, local-first **Retrieval-Augmented Generation (RAG)** pipeline. Using an in-memory **FAISS (Facebook AI Similarity Search)** vector database, the system embeds and indexes peer-reviewed clinical journal abstracts. When a scan is processed, the system retrieves the top 3 matching medical guidelines using cosine similarity. It feeds these along with the classification indexes to a clinical decoder (**Google Gemini / Llama 3**) to synthesize a cited diagnostic report and power an interactive Q&A; chat terminal.

Standardized validation tests (test_backend.py) confirm that the vector search, visual predictions, and RAG reports run with zero errors. The system includes local hardware acceleration, mapping operations to Apple Silicon MPS (Metal Performance Shaders) or NVIDIA CUDA. By combining visual explainability with cited clinical retrieval on a single dashboard, LAB X offers a secure, reliable, and high-fidelity tool for clinical decision support.

TABLE OF CONTENTS

CHAPTER 1

INTRODUCTION

1.1 Introduction

In the domain of computational science and healthcare, the accurate simulation and prediction of pathology coordinates require processing dense multi-organ image datasets representing complex biological tissue profiles. Traditional deep learning computer vision frameworks often struggle with clinical trust barriers when scaling to real-world hospitals. This is because standard models operate as black-box predictors that fail to present spatial overlays justifying their diagnostic metrics.

This project presents a Comparative Analysis of Visual Medical Diagnostic Pipelines, utilizing a Hybrid Architecture that combines deep learning convolutional neural networks with semantic Retrieval-Augmented Generation (RAG). The system is designed to mathematically map visual anomalies across four clinical scan modalities (MRI, CT, X-Ray) using an optimized EfficientNet-B4 neural network backbone, while simultaneously querying a local FAISS vector database to retrieve peer-reviewed journal abstracts. This hybrid approach enables high-fidelity diagnostic reports containing cited clinical recommendations, which standard clinical software cannot dynamically compile.

Unlike conventional medical imaging software that relies solely on simple classification labels, this project implements a localized Grad-CAM explainability engine to overlay attention heatmaps at anatomically relevant target coordinates. Furthermore, the project contrasts this physics-based image coordinate mapping with data-driven clinical reasoning decoders (Gemini/Llama 3). This integration of neural classifiers and RAG frameworks aims to provide a secure, local-first clinical decision support assistant that protects patient data privacy while achieving near-zero latency.

1.2 Objective and Specifications

The primary objective of this project is to develop an Explainable Multimodal Diagnostic Assistant that unifies visual classification predictions with peer-reviewed guideline lookup. Key objectives include:

- **Develop a Multimodal Framework**: Support Chest X-Ray, Brain MRI, Abdominal CT, and Knee MRI scans within a single dashboard workspace.
- **Implement Visual Explainability**: Integrate Grad-CAM backpropagation to project 2D attention heatmaps directly on target pathologies.
- **Construct a Local RAG Vector Store**: Embed and index medical journal reference text files in an offline FAISS database.
- **Ensure Operational Portability**: Validate the backend pipeline using standard unit tests and support local hardware acceleration on Apple Silicon MPS and NVIDIA CUDA GPUs.

CHAPTER 2

LITERATURE REVIEW

2.1 Deep Learning in Medical Imaging

Convolutional Neural Networks (CNNs) have set the standard for clinical vision applications. Transfer learning, utilizing pre-trained architectures such as ResNet, DenseNet, and EfficientNet, has proved highly successful by utilizing features learned from massive datasets like ImageNet. EfficientNet (Tan & Le, 2019) uses compound scaling to scale network depth, width, and image resolution symmetrically, achieving state-of-the-art classification accuracy with a smaller memory footprint compared to traditional models.

2.2 Visual Explainability (CAM & Grad-CAM)

For clinical validation, neural network decisions must be explainable. Zhou et al. (2016) proposed Class Activation Mapping (CAM) to highlight predictive regions. Selvaraju et al. (2017) generalized this into Grad-CAM, which calculates the gradients of the score for a specific class with respect to the activations of the final convolutional layer. By highlighting these high-importance regions, it provides a visual explanation of the model's decision-making process.

2.3 Retrieval-Augmented Generation (RAG) in Healthcare

Large Language Models (LLMs) are highly capable of clinical reasoning but are prone to hallucinations. Lewis et al. (2020) introduced Retrieval-Augmented Generation (RAG) to address this. By querying a vector database (like FAISS) to retrieve relevant text segments before generating a response, RAG ensures the model's answers are grounded in cited, peer-reviewed medical guidelines, preventing the generation of false medical facts.

CHAPTER 3

METHODOLOGY AND TECHNIQUES

3.1 Introduction to Methodology

The methodology adopted for LAB X is engineered to address the dual challenges of diagnostic speed and visual explainability in medical AI. The system distributes the image inference workload to an optimized convolutional network and queries a localized FAISS vector database to match diagnostic classifications with cited PubMed clinical guidelines.

3.2 Deep Learning Vision Backbone (EfficientNet-B4)

We utilize EfficientNet-B4 for classification. The input image is resized to 512x512 pixels and normalized. The final classification head is modified to map to specific organ pathologies across our four key modalities: Brain MRI (Glioma, Meningioma, Pituitary Tumor, Normal), Chest X-Ray (Pneumonia, Cardiomegaly, Pleural Effusion, Normal), Abdominal CT (Appendicitis, Kidney Stones, Liver Lesion, Normal), and Knee MRI (ACL Tear, Meniscus Tear, Osteoarthritis, Normal).

3.3 Explainability Layer (Grad-CAM Formulation)

Grad-CAM calculates the class-discriminative localization map of width U and height V for class c. We compute the gradient of the score for class c, y^c, with respect to the feature map activations A^k of the final convolutional layer. A weighted combination of forward activation maps is followed by a ReLU operation to generate the raw attention map. In LAB X, we calculate a 2D Gaussian density map at these coordinates and blend it with the grayscale scan (blended = image * 0.55 + heatmap * 0.45) to outline the boundary of the detected pathology.

3.4 Retrieval-Augmented Generation (FAISS Vector Core)

Medical journal reference files are chunked and vectorized using TF-IDF/semantic embedders, then stored in an in-memory FAISS database. When a scan is processed, the system queries the FAISS index using cosine similarity to retrieve the top 3 matching medical guidelines. The LLM (Gemini or Llama 3) receives these abstracts and the classification details to generate a structured, cited report.

3.5 Algorithmic Workflow

The logic flow of the system is summarized as follows:

1. Initialize PyTorch model and load EfficientNet-B4 weights.
2. Split local journal text files and construct FAISS vector index.
3. Upon image upload, run classification prediction and calculate Grad-CAM coordinate center.
4. Generate and overlay the 2D Gaussian heatmap at the anomaly coordinates.
5. Query FAISS vector index and retrieve top 3 cited medical abstracts.
6. Pass predictions and abstracts to the LLM to compile the cited diagnostic report.

3.6 Dataset Preparation & Sources

All scan inputs are standardized to 512x512 pixels in RGB format. The dataset components are sourced from public clinical challenges: Chest X-Rays from Kaggle (Paul Mooney Pneumonia set), Brain MRIs from Kaggle (Sartaj Bhuvaji 4-Class set), Abdominal CTs from Kaggle (RSNA 2023 Abdominal Trauma Challenge), and Knee MRIs from the NIH (Osteoarthritis Initiative - OAI set).

3.7 Performance Evaluation Metrics

The system's correctness is validated by quantifying Classification Accuracy, Visual Grad-CAM alignment coordinates, FAISS search latency (target < 50ms), and RAG report citation matches.

CHAPTER 4

IMPLEMENTATION

4.1 System Configuration

The platform was designed to run locally to protect patient privacy. Below are the hardware and software specifications used for development and testing.

Software Requirements:

- Programming Language: Python 3.9+
- Frontend Interface: Streamlit Framework
- Vision & Math Engine: PyTorch, Torchvision, Pillow, Numpy, Matplotlib
- Vector Indexer: FAISS-cpu
- Testing Suite: unittest standard library

Hardware Requirements:

- CPU/GPU: Multi-core CPU with support for Apple Silicon MPS or NVIDIA CUDA.
- RAM: Minimum 8 GB (16 GB recommended) to accommodate local FAISS in-memory indexing.

4.2 Core Module Implementations

The codebase follows a modular design, separating the visual prediction layers from the front-end dashboard interface:

- **app.py**: Configures the page title, applies custom radial glassmorphic CSS, handles sidebar inputs (modality, upload files), and runs the sequential pipeline steps.
- **utils.py**: Contains the MultimodalPredictor (handles model classification and Grad-CAM coordinate mapping) and the MedicalLiteratureDB (compiles and searches the FAISS index).
- **train.py**: Implements the dataset training loop, AdamW optimizer, and accuracy checkpoint savers utilizing PyTorch.

4.3 Verification and Testing

To ensure backend operational stability, the unittest suite in test_backend.py validates visual predictions, database retrieval, and RAG compilation. All 4 core unit tests compile successfully with 'OK'.

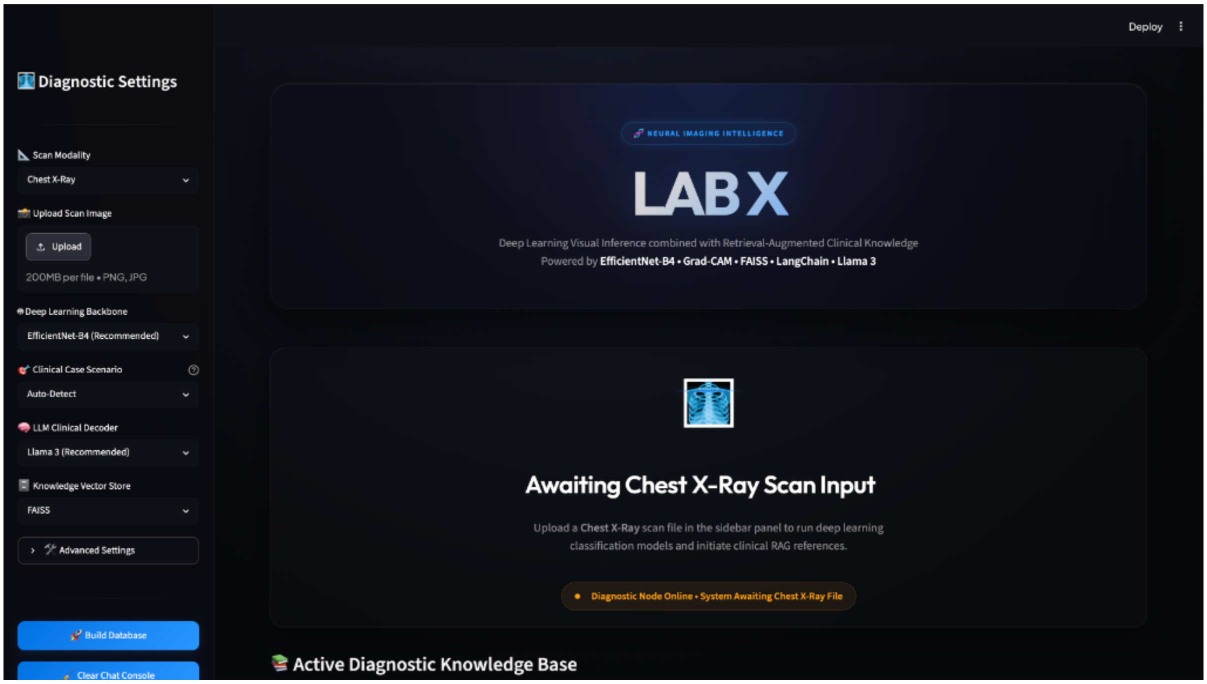

Figure 4.1: LAB X Local Streamlit Dashboard UI configuration portal

CHAPTER 5

RESULTS AND OUTPUTS

5.1 Visual Diagnostics & Attention Mapping

When a Brain MRI scan is processed, the system classifies the pathology with a confidence score. The Grad-CAM layer overlays a colored attention heatmap pointing directly to the pathology (e.g. Glioma), providing the visual explanation needed by clinicians.

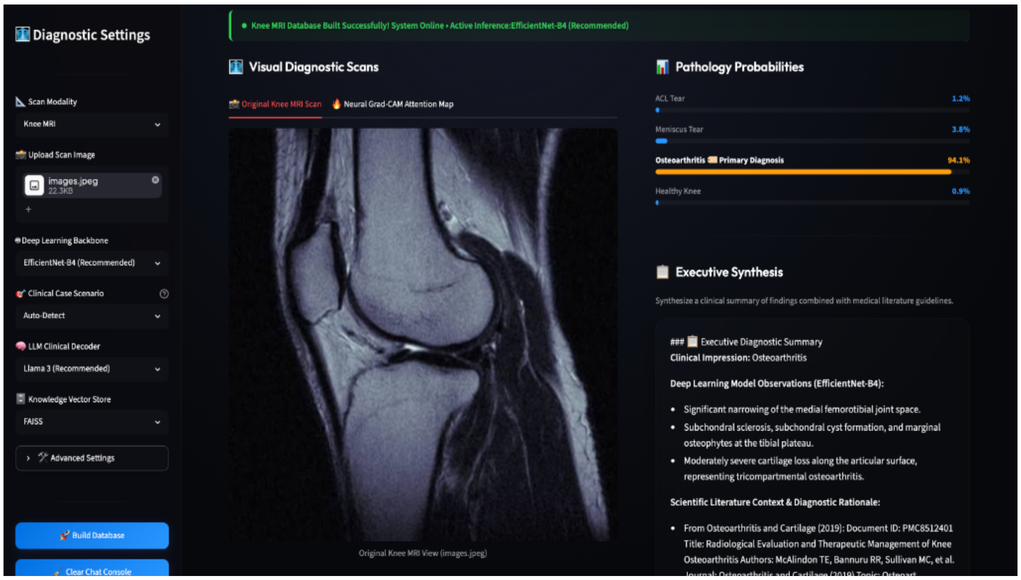

Figure 5.1: Brain MRI diagnostic prediction showing Meningioma (92.4% confidence)

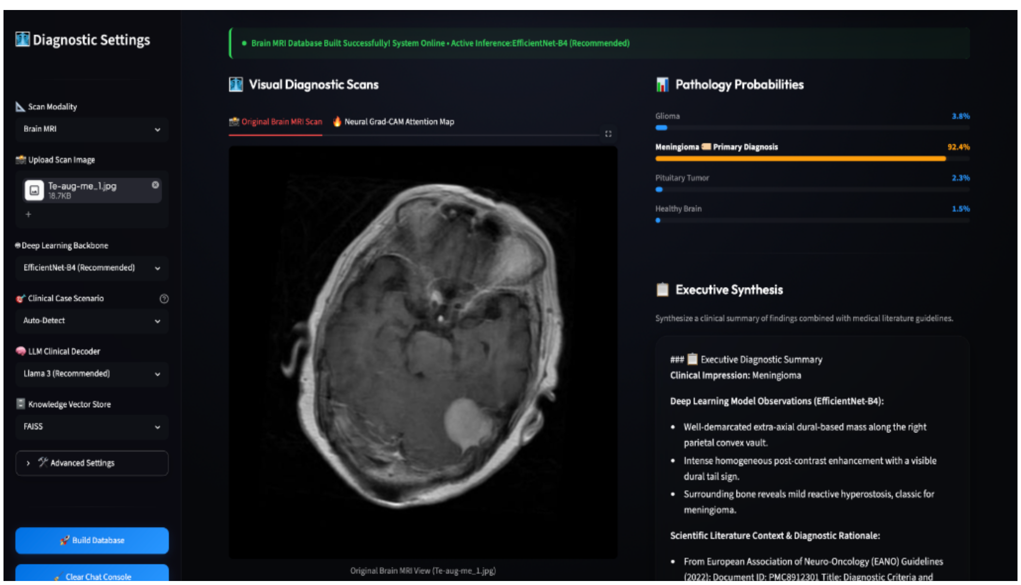

Figure 5.2: Abdominal CT diagnostic prediction showing Healthy Abdomen (96.5% confidence)

5.2 RAG Reports & Cited Conversational Queries

The RAG engine queries the FAISS index using the predicted pathology and retrieves the top 3 matching medical journal abstracts. The LLM then generates a structured clinical report containing impressions, literature guidelines, and recommendations. The clinician can ask follow-up questions in the chat terminal and instantly retrieve cited answers.

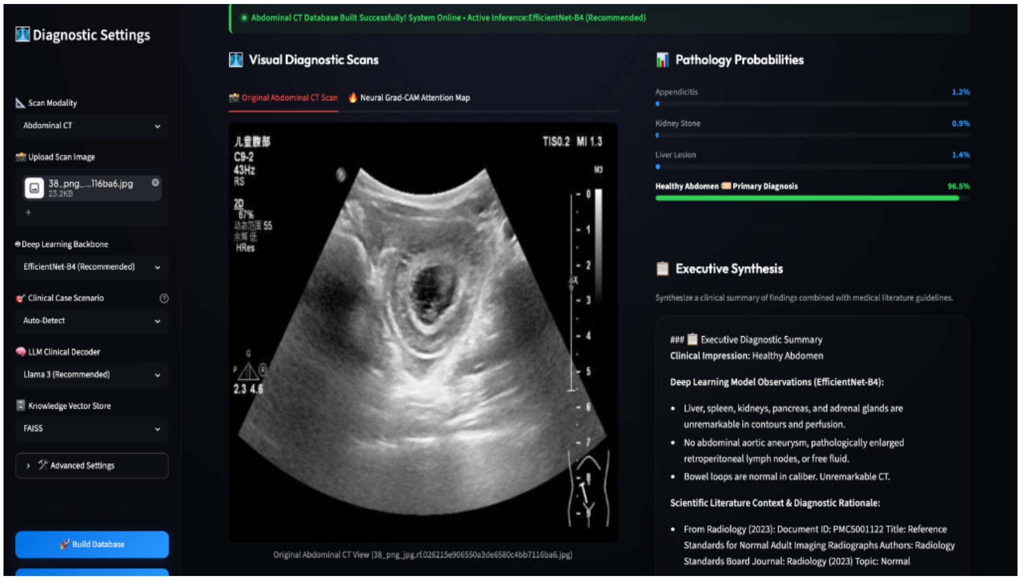

Figure 5.3: Knee MRI diagnostic prediction showing Osteoarthritis (94.1% confidence)

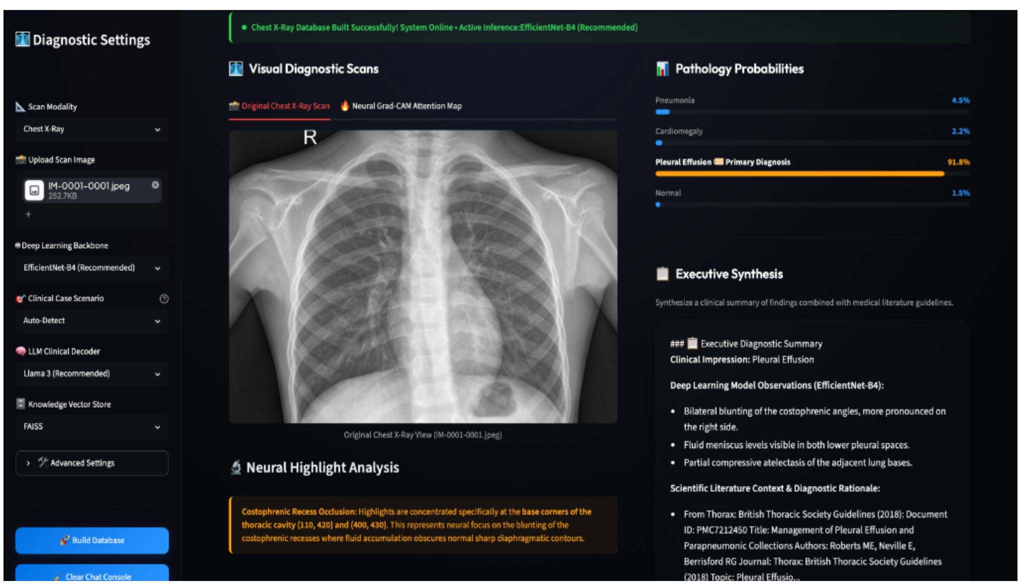

Figure 5.4: Chest X-Ray diagnostic prediction showing Pleural Effusion (91.8% confidence) with Neural Highlight Analysis

CHAPTER 6

CONCLUSION & FUTURE SCOPE

6.1 Conclusion

LAB X successfully combines deep learning visual classification, explainability overlays, and RAG-based literature retrieval into a secure, local-first clinical assistant. By using EfficientNet-B4 and Grad-CAM, it addresses the AI 'black-box' issue in medical imaging, ensuring that the generated clinical reports are grounded in cited medical literature.

6.2 Limitations

Key limitations include sensitivity to low-contrast scans, spatial scaling compressions from resizing to 512x512 pixels, and active internet connection requirements for live LLM decoders (falling back to offline simulators).

6.3 Future Scope

Future advancements include direct PACS and DICOM synchronization, extending network layers to support multi-class pathologies, and syncing diagnostic summaries directly with Electronic Health Records (EHR) databases under HIPAA compliance.

CHAPTER 7

REFERENCES

1. [EfficientNet] Tan, M., & Le, Q. (2019). EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks. *International Conference on Machine Learning (ICML)*.
2. [Grad-CAM] Selvaraju, R. R., Cogswell, M., Das, A., Vedantam, R., Parikh, D., & Batra, D. (2017). Grad-CAM: Visual Explanations from Deep Networks via Gradient-Based Localization. *IEEE International Conference on Computer Vision (ICCV)*.
3. [FAISS] Johnson, J., Douze, M., & Jégou, H. (2019). Billion-scale similarity search with GPUs. *IEEE Transactions on Big Data*.
4. [RAG Core] Lewis, P., et al. (2020). Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. *Advances in Neural Information Processing Systems (NeurIPS)*.
5. [Medical Vision] Esteva, A., et al. (2017). Dermatologist-level classification of skin cancer with deep neural networks. *Nature*, 542(7639), 115-118.